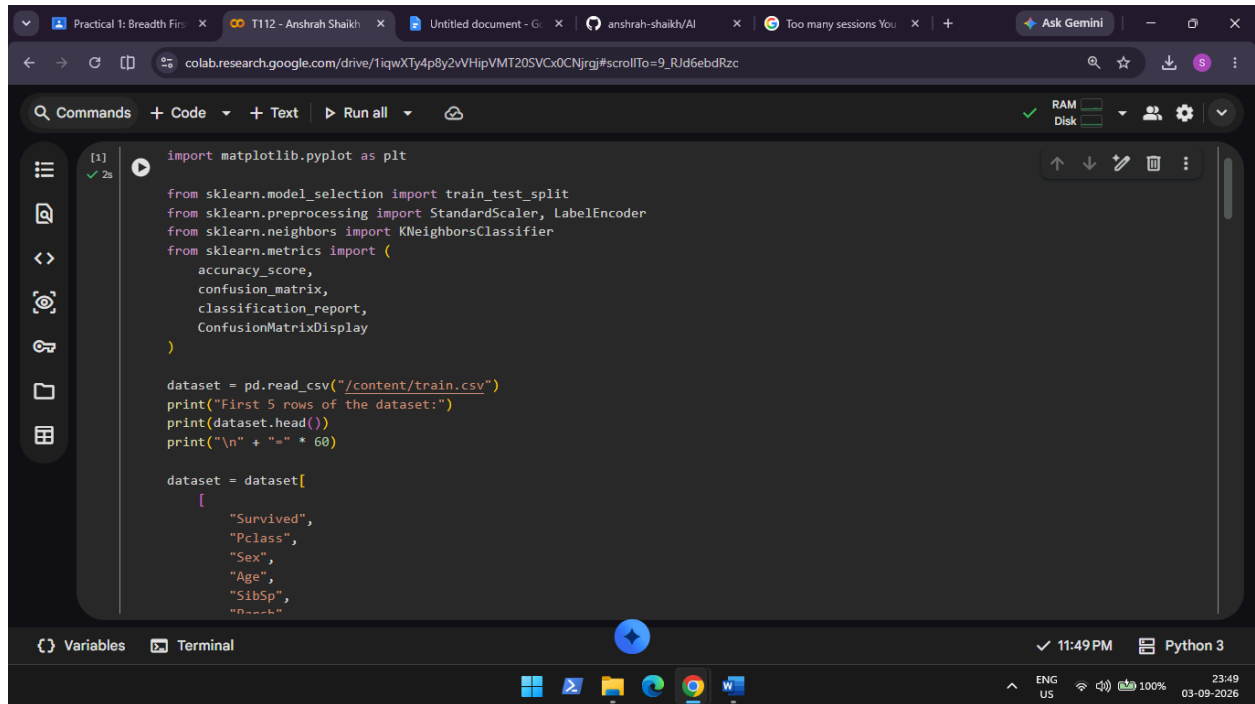


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AIM: K-Nearest Neighbors (K-NN)

- Implement the K-NN algorithm for classification or regression.
- Apply the K-NN algorithm to a given dataset and predict the class or value for test data.
- Evaluate the accuracy or error of the predictions and analyze the results.

INPUT:

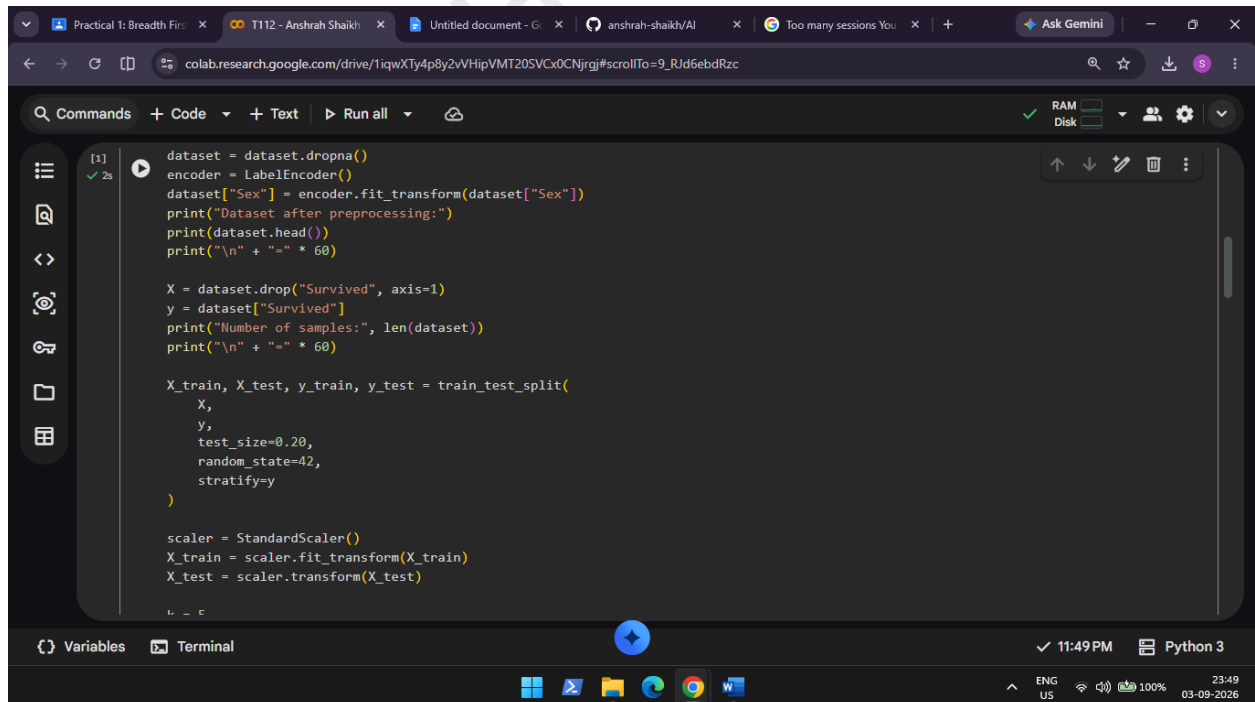


```
[1] import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import (
    accuracy_score,
    confusion_matrix,
    classification_report,
    ConfusionMatrixDisplay
)

dataset = pd.read_csv("/content/train.csv")
print("First 5 rows of the dataset:")
print(dataset.head())
print("\n" + "=" * 60)

dataset = dataset[
    [
        "Survived",
        "Pclass",
        "Sex",
        "Age",
        "SibSp",
        "PassengerId"
    ]
]
```



```
[1] dataset = dataset.dropna()
encoder = LabelEncoder()
dataset["Sex"] = encoder.fit_transform(dataset["Sex"])
print("Dataset after preprocessing:")
print(dataset.head())
print("\n" + "=" * 60)

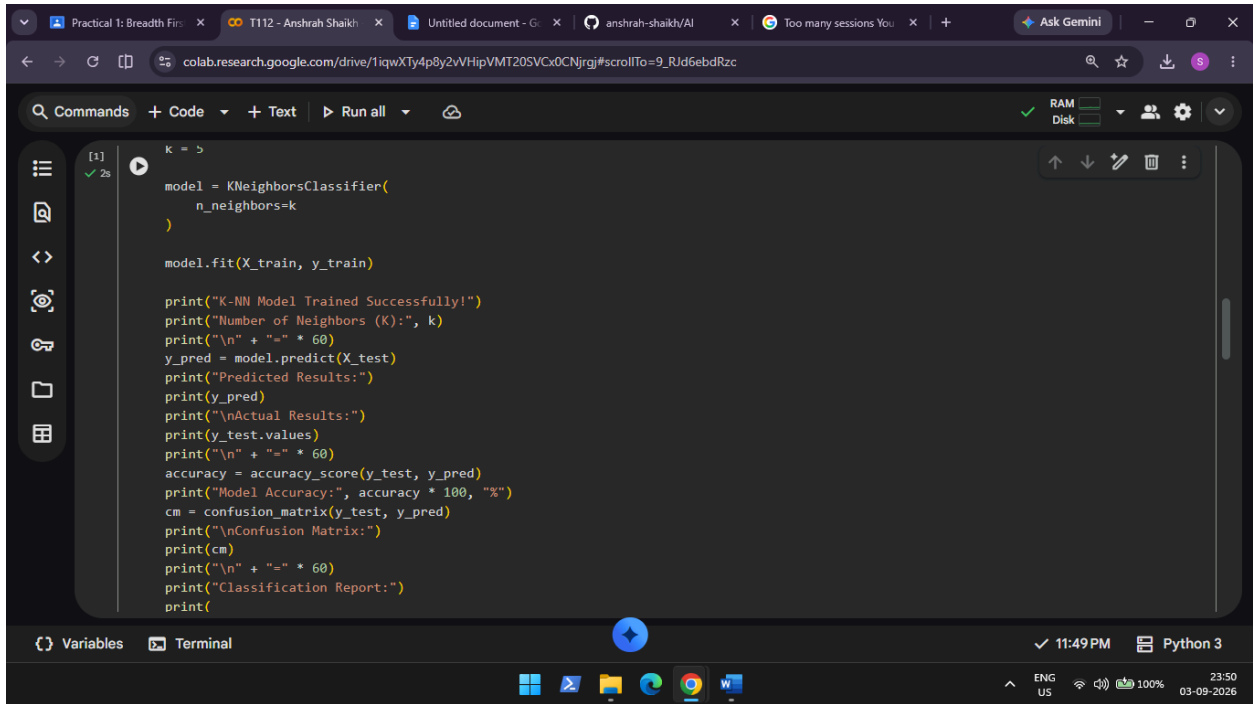
X = dataset.drop("Survived", axis=1)
y = dataset["Survived"]
print("Number of samples:", len(dataset))
print("\n" + "=" * 60)

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42,
    stratify=y
)

scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

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The screenshot shows a Google Colab notebook with the following code in the first cell:

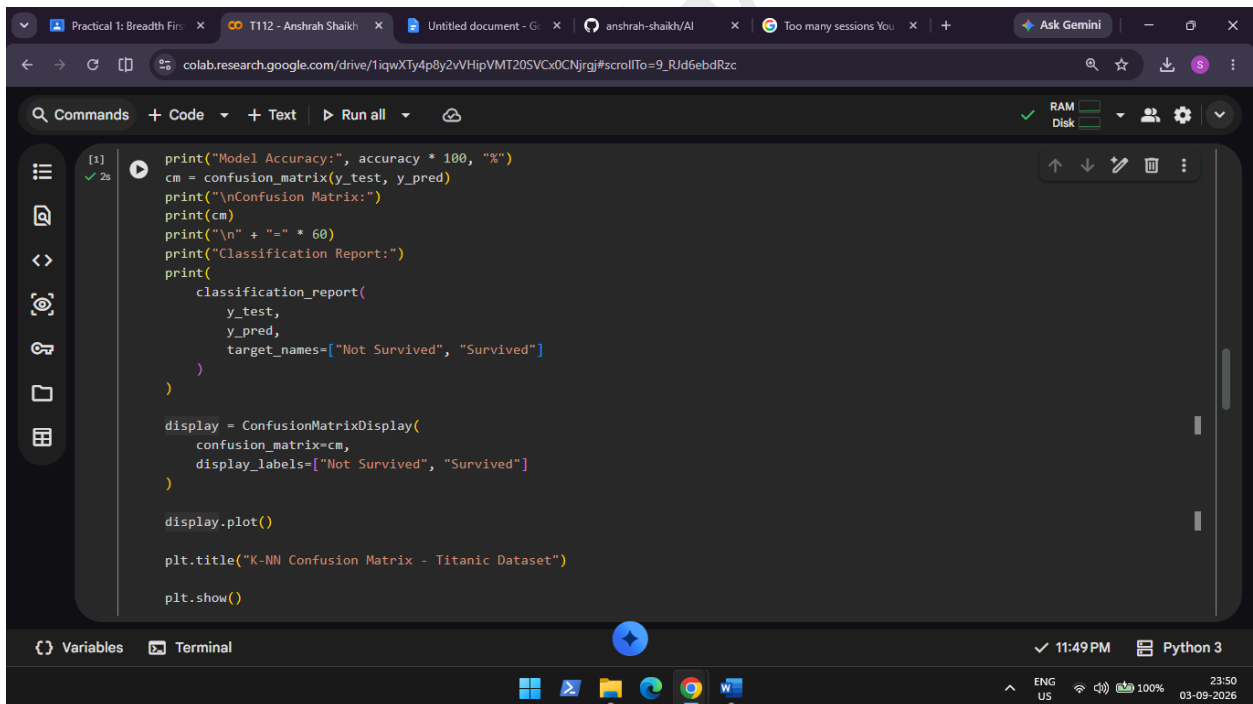
```
k = 5

model = KNeighborsClassifier(
    n_neighbors=k
)

model.fit(X_train, y_train)

print("K-NN Model Trained Successfully!")
print("Number of Neighbors (K):", k)
print("\n" + "=" * 60)
y_pred = model.predict(X_test)
print("Predicted Results:")
print(y_pred)
print("\nActual Results:")
print(y_test.values)
print("\n" + "=" * 60)
accuracy = accuracy_score(y_test, y_pred)
print("Model Accuracy:", accuracy * 100, "%")
cm = confusion_matrix(y_test, y_pred)
print("\nConfusion Matrix:")
print(cm)
print("\n" + "=" * 60)
print("Classification Report:")
print()
```

The interface includes a top bar with tabs for 'Practical 1: Breadth First Search', 'T112 - Anshrah Shaikh', and 'Untitled document - Google'. The bottom status bar shows '11:49 PM', 'Python 3', and system icons for language (ENG US), battery (100%), and date (03-09-2026).



The screenshot shows the second cell of the Google Colab notebook with the following code:

```
print("Model Accuracy:", accuracy * 100, "%")
cm = confusion_matrix(y_test, y_pred)
print("\nConfusion Matrix:")
print(cm)
print("\n" + "=" * 60)
print("Classification Report:")
print(
    classification_report(
        y_test,
        y_pred,
        target_names=["Not Survived", "Survived"]
    )
)

display = ConfusionMatrixDisplay(
    confusion_matrix=cm,
    display_labels=["Not Survived", "Survived"]
)

display.plot()

plt.title("K-NN Confusion Matrix - Titanic Dataset")

plt.show()
```

The interface is consistent with the first screenshot, showing the same top bar and bottom status bar.

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OUTPUT:

```
colab.research.google.com/drive/1iqwXTy4p8y2vHipVMT205VCx0CNjrgj#scrollTo=9_RJd6ebdRzc

Commands + Code + Text Run all

First 5 rows of the dataset:
  PassengerId  Survived  Pclass
0            1         0       3
1            2         1       1
2            3         1       3
3            4         1       1
4            5         0       3

   Name                               Sex  Age  SibSp
0  Braund, Mr. Owen Harris             male  22.0    1
1  Cumings, Mrs. John Bradley (Florence Briggs Th... female  38.0    1
2  Heikinen, Miss. Laina                female  26.0    0
3  Futrelle, Mrs. Jacques Heath (Lily May Peel)    female  35.0    1
4  Allen, Mr. William Henry             male  35.0    0

   Parch  Ticket   Fare Cabin Embarked
0      0   A/5 21171   7.2500   NaN      S
1      0   PC 17599  71.2833   C85      C
2      0  STON/O2. 3101282   7.9250   NaN      S
3      0  113803   53.1000  C123      S
4      0  373450   8.0500   NaN      S

Dataset after preprocessing:
  Survived  Pclass  Sex  Age  SibSp  Parch  Fare
0         0       3   1  22.0    1      0   7.2500
```

```
colab.research.google.com/drive/1iqwXTy4p8y2vHipVMT205VCx0CNjrgj#scrollTo=9_RJd6ebdRzc

Commands + Code + Text Run all

0      0       3   1  22.0    1      0   7.2500
1      1       1   0  38.0    1      0  71.2833
2      1       3   0  26.0    0      0   7.9250
3      1       1   0  35.0    1      0  53.1000
4      0       3   1  35.0    0      0   8.0500

Number of samples: 714

K-NN Model Trained Successfully!
Number of Neighbors (K): 5

Predicted Results:
[0 1 1 0 0 0 0 0 1 1 0 0 0 0 0 0 0 0 0 1 1 0 1 1 0 1 1 0 1 1 0 1 1
 0 1 0 1 0 1 1 0 0 1 0 1 1 1 0 1 0 0 0 0 0 0 0 1 1 0 0 0 0 1 1 1 1 0 1 0
 0 1 1 1 0 1 0 0 1 0 0 0 0 1 1 0 0 0 0 0 0 0 1 0 0 1 0 0 0 0 0 1 1 0 0 0
 0 0 1 0 1 0 1 0 0 0 1 1 0 0 0 0 1 1 0 1 0 1 1 1 1 0 0 0 0 1 0 0 0]

Actual Results:
[1 1 0 0 1 0 0 0 0 1 0 0 1 0 0 0 0 0 0 0 1 1 0 1 1 0 1 1 0 1 1 0 1 1
 1 0 0 0 0 0 1 0 0 1 0 1 1 1 0 1 0 0 1 0 0 0 0 1 1 0 0 0 0 1 1 0 1 1 0 0
 0 0 1 0 0 1 0 0 1 1 1 0 0 1 1 0 1 0 0 0 0 0 0 0 1 1 0 1 0 0 0 0 1 1 1 0
 0 0 0 1 1 1 1 0 1 1 1 0 0 0 0 1 0 0 1 0 0 0 1 1 0 0 0 1 0 0 0]
```

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